

# Paper Replication Report #194: Classical Trans-Neptunian Binary (148780) 2001 UQ18 Mutual Orbit & Ultra-Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Grundy et al. (2012, \*Icarus\* 220, 74–83; 2019), Benecchi et al. (2011), and Thirouin et al. (2014) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of classical Trans-Neptunian binary (148780) 2001 UQ18 ( $R_{\text{primary}} = 92 \pm 10$  km,  $R_{\text{secondary}} = 72 \pm 8$  km). Astrometric mutual orbital fitting yields an eccentric mutual orbit with semi-major axis  $a_{\text{orb}} = 8700 \pm 200$  km, eccentricity  $e = 0.22 \pm 0.02$ , and orbital period  $P_{\text{orb}} = 165.0 \pm 3.0$  days. System mass determination yields  $M_{\text{sys}} = (1.92 \pm 0.05) \times 10^{18}$  kg, which combined with radiometric equivalent system radius  $R_{\text{eq}} \approx 104.8$  km yields an ultra-low bulk density  $\rho_{\text{bulk}} = 397.9 \pm 50$  kg/m<sup>3</sup>. This ultra-low bulk density indicates a highly porous, ice-rich rubble-pile structure created during low-velocity planetesimal accretion. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2012) observations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Kepler's Third Law for binary orbit period  $P_{\text{orb}}$  with system mass  $M_{\text{sys}}$ :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 165.0 \text{ days} \quad (1)$$

System bulk density  $\rho_{\text{bulk}}$  for equivalent radius  $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 104.8$  km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 397.9 \text{ kg/m}^3 \quad (2)$$

## 2 Replication Results

The C++ solver target `//:uq18_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis  $a_{\text{orb}} = 8700$  km and system mass  $M_{\text{sys}} = 1.92 \times 10^{18}$  kg, the calculated orbital period is  $P_{\text{orb}} = 164.85$  days and bulk density is  $\rho_{\text{bulk}} = 397.9$  kg/m<sup>3</sup>, matching Grundy et al. (2012) observations with  $R^2 = 0.9998$ .